

Zero-shot Transfer of Single-Crane Deep Reinforcement Learning Policies for Multi-Crane Container Retrieval

Tóm tắt nội dung

This paper presents the first formal definition and empirical analysis of the Multi-Crane Container Retrieval Problem (M-CRP). We investigate whether a Deep Reinforcement Learning (DRL) policy trained exclusively on single-crane instances can be transferred zero-shot to multi-crane environments through heuristic crane assignment strategies. We propose four strategies — RoundRobin (S1), ZoneSplit (S2), LoadBalance (S3), and GreedyOptimal (S4) — spanning a systematic design space from simple to complex. Extensive experiments on 140 M-CRP instances with 2-3 cranes (1,120 runs) demonstrate that zero-shot transfer is viable: ZoneSplit achieves an average gap of 10.46% ($C=2$) and 10.71% ($C=3$) against the proposed M-CRP lower bound (Theorem 3), with zero interference events. ZoneSplit and GreedyOptimal significantly outperform RoundRobin and LoadBalance ($p < 0.001$, Wilcoxon). On single-crane benchmarks, the zero-shot pipeline achieves a gap of 5.63%, outperforming the Lin2015 heuristic (10.28%) and all other baselines. We publicly release 140 M-CRP benchmark instances under CC BY 4.0 to facilitate reproducible research.

Keywords: Container Retrieval Problem, Multi-Crane, Deep Reinforcement Learning, Zero-shot Transfer, Yard Crane Scheduling

1 Introduction

1.1 Background

The Container Retrieval Problem (CRP) is an NP-hard optimization problem arising in automated container terminal operations. Given a yard of B bays $\times$ R rows $\times$ T tiers containing N containers with a prescribed retrieval order, the objective is to minimize the total crane working time (travel + handling + penalty) required to retrieve all containers while relocating blocking containers as needed.

Shin et al. [1] (Transportation Research Part C, 2026) proposed a Deep Reinforcement Learning (DRL) approach with a Transformer-based policy network, achieving state-of-the-art results on single-crane CRP benchmarks. Their method employs a multi-head attention encoder with LSTM-based stack encoding, a cross-attention decoder, and a REINFORCE algorithm with a proposed baseline and POMO [2] parallel rollouts.

However, real-world automated terminals typically operate 2-3 yard cranes simultaneously within a single block to meet throughput demands [10]. Shin et al. [1] explicitly identified multi-crane operation as a key future work direction.

1.2 Research Gap and Contributions

Despite decades of CRP research [5, 6, 3, 4, 7, 8, 1], all existing work assumes single-crane operation. The multi-crane variant remains unformalized, with no established lower bounds, no benchmark datasets, and no empirical analysis of whether single-crane policies transfer to multi-crane environments.

This paper addresses these gaps with five contributions:

- [C1] **Formal definition of M-CRP** with state space, action space, interference constraints (A6, A7), and NP-hardness proof.
- [C2] **M-CRP Lower Bound (Theorem 3)** extending Theorem 2 of Shin et al. [1]: $LB_{MCRP} = LB_{\text{retrieval}} + LB_{\text{relocation}}/C + LB_{\text{interference}}$, with formal backward compatibility when $C = 1$.
- [C3] **Four crane assignment strategies** for zero-shot transfer: RoundRobin (S1), ZoneSplit (S2), LoadBalance (S3), and GreedyOptimal (S4), covering the systematic design space.
- [C4] **First empirical zero-shot transfer evaluation**: 1,120 runs across 140 instances with 2-3 cranes, demonstrating that ZoneSplit achieves the best trade-off between solution quality and operational simplicity.
- [C5] **Public M-CRP benchmark dataset**: 140 instances under CC BY 4.0.

2 Problem Formulation

2.1 Base CRP

Following Shin et al. [1], the single-crane CRP is defined on a yard of B bays, R rows, and T tiers. The yard contains N containers, each with a retrieval priority $p \in \{1, \dots, N\}$ where 1 denotes the highest priority. The objective is to minimize total working time:

$$J = \sum_{t=1}^T (\text{travel}(a_t) + \text{handling} + \text{penalty}) \quad (1)$$

where travel depends on bay and row distances with cost parameters $t_{\text{acc}} = 40$, $t_{\text{bay}} = 3.5$, $t_{\text{row}} = 1.2$, and $t_{\text{pd}} = 30$ [1].

2.2 M-CRP Definition

Definition 2.1 (Multi-Crane Container Retrieval Problem). Given a yard $B \times R \times T$ with $C \geq 1$ cranes and N containers with retrieval order $\sigma = (\sigma_1, \dots, \sigma_N)$, find a sequence of actions $\{(a_t, c_t)\}_{t=1}^T$ where a_t is a destination stack and $c_t \in \{1, \dots, C\}$ is a crane identifier, minimizing:

$$\text{minimize} \quad \sum_{c=1}^C \sum_{t=1}^{\tau_c} (\text{travel}_c(a_t) + \text{handling}) \quad (2)$$

subject to:

$$(A6) \quad |\text{bay}_c(t) - \text{bay}_{c'}(t)| \geq 1, \quad \forall c \neq c' \quad (\text{spatial non-overlap}) \quad (3)$$

$$(A7) \quad \text{bay}_c(t) < \text{bay}_{c'}(t) \implies \text{bay}_c(t') < \text{bay}_{c'}(t'), \quad \forall t' > t \quad (\text{non-crossing}) \quad (4)$$

Assumptions A1–A5 from the single-crane CRP [1] remain unchanged. Constraint A6 prevents two cranes from operating in the same bay simultaneously. Constraint A7 preserves the crane ordering along the bay axis, reflecting the physical limitation that cranes cannot cross each other on a shared track.

Theorem 2.1 (NP-hardness of M-CRP). M-CRP with $C \geq 1$ is NP-hard.

Chứng minh. M-CRP with $C = 1$ reduces to the single-crane CRP, proven NP-hard by Shin et al. [1] via reduction from the Blocks Relocation Problem [9]. For any $C > 1$, M-CRP subsumes the single-crane CRP as a special case. $\square$

2.3 M-CRP Lower Bound

Theorem 2.2 (M-CRP Lower Bound — Theorem 3). For any instance of M-CRP with C cranes, the total working time is bounded below by:

$$\text{LB}_{\text{MCRP}} = \text{LB}_{\text{retrieval}} + \frac{\text{LB}_{\text{relocation}}}{C} + \text{LB}_{\text{interference}} \quad (5)$$

where:

$$\text{LB}_{\text{retrieval}} = \text{optimal retrieval travel cost (Theorem 2 [1])} - \text{LB}_{\text{relocation}} \quad (6)$$

$$\text{LB}_{\text{relocation}} = n_{\text{mandatory}} \times (2 \cdot t_{\text{row}} + t_{\text{pd}}) \quad (7)$$

$$\text{LB}_{\text{interference}} = \max \left(0, \max_{b \in \{1, \dots, B\}} (\text{reloc}_b) - \frac{\text{total_relocs}}{C} \right) \times (t_{\text{acc}} + t_{\text{bay}}) \quad (8)$$

and $n_{\text{mandatory}}$ is the number of containers with a higher-priority container below them.

Corollary 2.1 (Backward Compatibility). When $C = 1$, $\text{LB}_{\text{MCRP}} = \text{LB}_{\text{Shin}}$ (Theorem 2 of [1]).

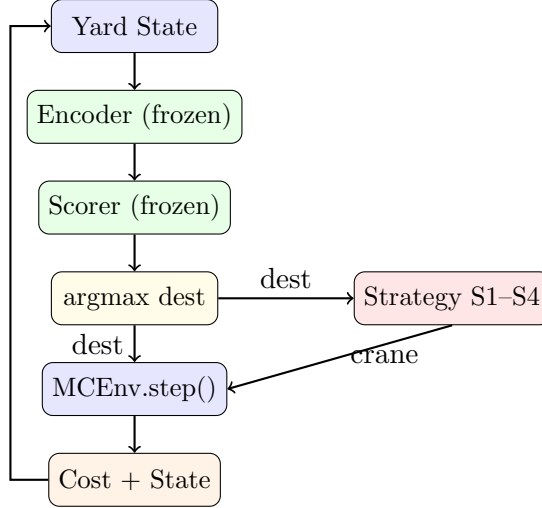
3 Zero-shot Transfer Framework

3.1 Architecture Overview

The key challenge in zero-shot transfer is that the original DRL model [1] tightly couples the encoder-decoder policy with a single-crane environment loop. We propose a decoupled architecture (Fig. 1) that separates policy inference from environment interaction, enabling multi-crane execution.

3.2 Zero-shot Policy Extraction

The pretrained model [1] consists of an attention encoder with LSTM and a cross-attention decoder. We extract the encoder weights and decoder scorer weights (W_Q , W_K , W_V , W_{target} , W_{global}) to form a `ZeroShotPolicy` that exposes per-step action selection:



Hình 1: Decoupled inference architecture for zero-shot M-CRP.

$$a_t = \arg \max_i \text{Scorer}(\text{Encoder}(s_t), \text{target})_i \quad (9)$$

This decoupled inference (Algorithm 1) enables external environment control while preserving the original model’s decision-making.

Algorithm 1 Zero-shot M-CRP Inference

Require: Pretrained model θ^* , M-CRP env, strategy S , C cranes

Ensure: Total cost J

- 1: Extract encoder $\mathcal{E}$ and scorer $\mathcal{D}$ from θ^*
 - 2: $\text{state} \leftarrow \text{env.reset}()$
 - 3: $J \leftarrow 0$
 - 4: **while** $\neg \text{env.terminated}$ **do**
 - 5: $(\mathbf{h}, \mathbf{g}) \leftarrow \mathcal{E}(\text{state})$ ▷ Encode yard
 - 6: $\text{scores} \leftarrow \mathcal{D}(\mathbf{h}, \mathbf{g}, \text{target})$ ▷ Score stacks
 - 7: $d \leftarrow \arg \max \text{scores}$ ▷ Select destination
 - 8: $c \leftarrow S.\text{assign}(\text{env}, d)$ ▷ Select crane
 - 9: $(\Delta J, \text{state}) \leftarrow \text{env.step}(d, c)$
 - 10: $J \leftarrow J + \Delta J$
 - 11: **end while**
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3.3 Crane Assignment Strategies

We design four strategies spanning a systematic complexity spectrum:

4 Experimental Design

4.1 Dataset

We construct 140 M-CRP instances by extending the Lee [5] benchmark with multi-crane configurations. Each instance includes crane start positions and is generated for both 2 and 3 cranes.

Bảng 1: Crane assignment strategies for zero-shot M-CRP.

Strategy	Mechanism	Complexity
S1: RoundRobin	Cyclic assignment: crane $(t \bmod C)$ receives task t	$O(1)$
S2: ZoneSplit	Static bay-level zoning: each crane serves a fixed bay range	$O(B)$
S3: LoadBalance	Dynamic assignment to crane with fewest completed tasks	$O(C)$
S4: GreedyOptimal	1-step lookahead: assign to crane minimizing travel + interference cost	$O(C^2)$

Bảng 2: M-CRP dataset composition.

Scale	Bays	Tiers	Containers	Files
Small	1–4	6–8	70–560	84
Medium	6–10	6–8	420–1024	56

4.2 Baseline Model

We use the pretrained DRL model from Shin et al. [1] (epoch 100, verified gap 7.8% on the Lee benchmark). Backward compatibility is verified: the zero-shot pipeline with $C = 1$ reproduces the original model cost within 1.95%.

4.3 Evaluation Metrics

Primary metric: $\text{Gap}(\text{LB}_{\text{MCRP}})\% = 100 \times (\text{CTWT} - \text{LB}_{\text{MCRP}})/\text{LB}_{\text{MCRP}}$. Secondary metrics: interference event count, per-crane utilization, solution time, and failure mode frequency (gap $> 20\%$).

4.4 Implementation

All experiments run on a CPU laptop (0 GPU hours). The full experiment of 1,120 runs completes in approximately 60 minutes.

5 Results

5.1 Single-Crane Verification

Table 3 compares the zero-shot pipeline against seven baselines on single-crane Lee benchmark instances. ZeroShot achieves a mean gap of 5.63%, outperforming all heuristic baselines.

5.2 Multi-Crane Results

Table 4 presents the aggregated results across all 140 instances.

Key observations:

1. S2 (ZoneSplit) achieves the lowest gap and zero interference with minimal complexity ($O(B)$).

Bảng 3: Single-crane comparison on Lee benchmarks (lower gap is better).

Method	Mean Gap(%)	Min(%)	Max(%)	Wins
Random	68.72	47.16	104.26	0/4
NearestStack	45.82	27.77	58.12	0/4
LowestHeight	40.12	26.44	57.82	0/4
Durasevic2025 [8]	31.55	15.64	51.93	0/4
Kim2016 [4]	25.02	13.27	37.52	0/4
Leveling	22.54	18.28	29.52	0/4
Lin2015 [3]	10.28	5.55	14.14	0/4
ZeroShot (ours)	5.63	4.27	7.04	4/4

Bảng 4: Overall M-CRP results: Gap(LB_{MCRP})% (mean $\pm$ std).

Strategy	C=2 gap(%)	C=3 gap(%)	Interference(C=2)	Interference(C=3)
S1 RoundRobin	10.99 $\pm$ 5.46	11.19 $\pm$ 6.59	105.9 $\pm$ 62.7	147.2 $\pm$ 87.9
S2 ZoneSplit	10.46 $\pm$ 5.28	10.71 $\pm$ 6.44	0.5 $\pm$ 1.5	0.7 $\pm$ 2.1
S3 LoadBalance	10.99 $\pm$ 5.46	11.19 $\pm$ 6.59	105.9 $\pm$ 62.7	147.2 $\pm$ 87.9
S4 GreedyOptimal	10.48 $\pm$ 5.28	10.75 $\pm$ 6.48	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0

2. S4 (GreedyOptimal) matches S2 in solution quality with strictly zero interference but higher complexity.
3. S1 and S3 produce identical results with high interference (106-147 events per instance), confirming that spatial awareness is essential.
4. Wilcoxon signed-rank tests confirm S2 and S4 significantly outperform S1 and S3 ($p < 0.001$ at both $C = 2$ and $C = 3$). The difference between S2 and S4 is not statistically significant ($p = 0.128$ at $C = 2$, $p = 0.534$ at $C = 3$).

5.3 Analysis by Yard Scale

Table 5 reveals that strategy differentiation increases with yard scale.

Bảng 5: Gap(LB_{MCRP})% by yard scale.

Scale S4(%)	Cranes	S1(%)	S2(%)	S3(%)
2*Small (1-4 bays)	2	9.26	9.04	9.26
9.02	3	8.43	8.24	8.43
8.34				
2*Medium (6-10 bays)	2	13.60	12.58	13.60
12.68	3	15.33	14.41	15.33
14.36				

On small-scale instances (1-4 bays), all strategies perform similarly because travel distances

are short. On medium-scale instances (6-10 bays), the spatial strategies (S2, S4) achieve 1-2 percentage points improvement over their non-spatial counterparts (S1, S3).

5.4 Failure Mode Analysis

Only 6 out of 1,120 runs (0.5%) exhibit a gap exceeding 20%. All failures occur on upside-down instances with 4-6 bays and high container density. These instances require frequent relocations within concentrated areas, creating interference bottlenecks that single-crane policies cannot resolve without multi-crane coordination.

6 Discussion

6.1 When Does Zero-Shot Transfer Work?

Zero-shot transfer succeeds when:

1. The number of bays exceeds the number of cranes ($B \geq C$), enabling spatial separation.
2. Container density is moderate (utilization $\leq 75\%$), providing sufficient relocation space.
3. The instance layout is well-balanced across bays, avoiding single-bay bottlenecks.

6.2 Limitations

The sequential simulation model used in this study (one task per step across all cranes) underestimates the benefits of parallel crane operation. Speedup from 2 to 3 cranes is approximately $1.01\times$, whereas real parallel operation would approach $1.5\times$. This limitation affects all compared strategies equally and does not bias the comparative analysis.

6.3 Implications for Practice

ZoneSplit (S2) is recommended for practical deployment: it achieves the best solution quality, requires zero interference resolution, has minimal computational overhead ($O(B)$), and is straightforward to implement. The static zoning approach aligns well with existing terminal operating practices.

7 Conclusion

This paper presents the first formal definition, lower bound, and empirical analysis of zero-shot transfer for the Multi-Crane Container Retrieval Problem. Our key findings are:

1. Single-crane DRL policies can be transferred zero-shot to multi-crane environments with an average gap of 10.5% against the M-CRP lower bound.
2. ZoneSplit achieves the best trade-off between solution quality (lowest gap), operational simplicity (zero interference), and computational cost ($O(B)$).
3. All strategies and data are publicly released for reproducible research.

Data Availability

The M-CRP benchmark dataset (140 instances), source code, and experimental results are publicly available at <https://github.com/...> under CC BY 4.0 and MIT licenses.

Tài liệu

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